

- [Home](#)
- [General Usage](#)
- [Term Search](#)
- [Ontology Property Search](#)
- [Annotator](#)
- [Recommender](#)
- [Batch](#)
- [Ontology Analytics](#)
- [Resources](#)
 - [Media Types and Hypermedia Links](#)
 - [Category](#)
 - [Class](#)
 - [Group](#)
 - [Instance](#)
 - [Mapping](#)
 - [Metric](#)
 - [Note](#)
 - [Reply](#)
 - [ObjectProperty](#)
 - [Ontology](#)
 - [OntologySubmission](#)
 - [Project](#)
 - [ProvisionalClass](#)
 - [ProvisionalRelation](#)
 - [Review](#)
 - [Collection](#)
 - [Scheme](#)
 - [Slice](#)
 - [User](#)
- [Content Types](#)

API Documentation

General Usage

This API is comprised of a set of resources (Ontologies, Classes, etc) and related endpoints (Search, Annotator, Recommender) that are connected together via links, much like webpages. We recommend that you try browsing the API using a web browser (Chrome and Firefox work very well while IE does not) before you start writing code. For more information, please see the documentation on [Media Types and Hypermedia Links](#) or view our [sample code](#), available in Java, Python, Ruby and other languages (please email support@bioontology.org if you would like examples in another language).

Common Parameters

Parameter	Possible Values	Description
apikey	{your api key}	An API Key is required to access any API call. It can be provided in three ways: 1. Using the apikey query string parameter 2. Providing an Authorization header: Authorization: apikey token=your_apikey (replace 'your_apikey' with your actual key) 3. When using a web browser to explore the API, if you provide your API Key once using method 1, it will be stored in a cookie for subsequent requests. You can override this by providing a different API Key in a new call.
include	all {comma-separated list of attributes, EX: attr1,attr2}	By default, the API will show a subset of the available attributes for a given media type. This behavior can be overridden by providing <code>include=all</code> to show all attributes or <code>include=attribute1,attribute2</code> to include a specific list. The API is optimized to return the default values, so overriding this can impact the performance of your request. The <code>include=all</code> option is most useful for testing in the browser. Use it to identify the set of attributes required and use only those by passing them as a comma separated list, e.g. <code>include=prefLabel,cui</code>. The <code>include</code> parameter is currently unsupported on Annotator and Recommender endpoints.
format	json jsonp xml	The API returns JSON as the default content type. This can be overridden by using the <code>format</code> query string parameter. The API also respects Accept header entries, with precedence given to the <code>format</code> parameter.
page	{integer representing the page number}	For calls that are paged, this will indicate which page number you want to retrieve. You can follow hypermedia links for <code>nextPage</code> and <code>prevPage</code> as well.
pagesize	{integer representing the size of the returned page}	For calls that are paged, this will indicate the size of the page you want to retrieve.
include_views	{boolean representing whether or not to include ontology views (default is false)}	For calls that involve ontologies, <code>include_views='true'</code> will include ontology views.

display_context	{true false} (defaults to true)	Turn off the JSON-LD context serialization. This will reduce the response size significantly for some calls, speeding up transmission and parse time.
display_links	{true false} (defaults to true)	Turn off the hypermedia link serialization. This will reduce the response size significantly for some calls, speeding up transmission and parse time.
download_format	{csv rdf} (defaults to user-uploaded file format)	Allows you to specify alternative formats for ontology file downloads for ontology and ontology submission download endpoints. CSV is only available for the most recent submission.

Search, Annotator, Recommender, and Resource Index Endpoints

Several endpoints are available for performing lookups for classes, annotations, and annotated resources. These endpoints are not strictly RESTful, but do return objects with links where relevant.

Term Search

An endpoint for searching terms and their properties is available

- Perform search
 - GET POST /search?q={search query}
 - example: [/search?q=melanoma](#)
- Parameters
 - ontologies={ontology_id1,ontology_id2,ontology_id3}
 - require_exact_match={true|false} // default = false
 - suggest={true|false} // default = false. Will perform a search specifically geared towards type-ahead suggestions.
 - also_search_views={true|false} // Include ontology views in the search. default = false
 - require_definitions={true|false} // default = false
 - also_search_properties={true|false} // default = false
 - also_search_obsolete={true|false} // default = false (exclude obsolete terms)
 - cui={C0018787,C0225807,C0018787} // Allows restricting query by CUIs. default = null (no restriction)
 - semantic_types={T023,T185,T061} // Allows restricting query by Semantic Types (TUI). default = null (no restriction)
 - include={prefLabel, synonym, definition, notation, cui, semanticType} // default = (see Common Parameters section)
 - page={integer representing the page number} // default = 1
 - pagesize={integer representing the size of the returned page} // default = 50
 - language={an ISO 639-1 language value, e.g 'fr' or 'en'} // by default search in English

Subtree Search

The same endpoint also allows limiting a search to a given subtree/branch. The root of the subtree is defined using a combination of parameters "ontology" and "subtree_root_id". Both are required when performing a subtree search.

- Perform a subtree search
 - example: [/search?q=melanoma&ontology=NCIT&subtree_root_id=http%3a%2f%2fncicb.nci.nih.gov%2fxml%2fowl%2fEVS%2fThesaurus.owl%23C3224](#)
- Parameters
 - ontology={ontology_id}
 - subtree_root_id={a URI encoded class id of the root node of the subtree}

Roots Only Search

The same endpoint allows limiting a search to root classes only.

- Perform a roots-only search
 - example: [/search?q=melanoma&ontology=NCIT.GO&roots_only=true](#)
- Parameters
 - roots_only={true|false}

Ontology Property Search

An endpoint for searching ontology properties by their labels and IDs (URIs)

- Perform search
 - GET POST /property_search?q={search query}
 - example: [/property_search?q=has dimension](#)
- Parameters
 - ontologies={ontology_id1,ontology_id2,ontology_id3}
 - require_exact_match={true|false} // Restricts results only to the exact matches of the query by the property id, label, or the generated label (a label, auto-generated from the ID). default = false
 - also_search_views={true|false} // Include ontology views in the search. default = false
 - require_definitions={true|false} // Filter results only to those that include definitions. default = false
 - include={label, labelGenerated, definition, parents} // default = All of the properties in this example
 - ontology_types={ONTOLOGY,VALUE_SET_COLLECTION} // default = all ontology types are returned
 - property_types={object,annotation,datatype} // Restricts results to specific property types (object, annotation, datatype). default = all property types are returned
 - page={integer representing the page number} // default = 1
 - pagesize={integer representing the size of the returned page} // default = 50

Annotator

The Annotator endpoint will examine text input and return relevant classes.

- Annotate text
 - GET POST /annotator?text={input text}
 - example: [/annotator?text=Melanoma+is+a+malignant+tumor+of+melanocytes+which+are+found+predominantly+in+skin+but+also+in+the+bowel+and+the+eye.](#)
- Parameters
 - Filtering & query behavior
 - ontologies={ontology_id1,ontology_id2...,ontology_idN}
 - semantic_types={semType1,semType2...,semTypeN}
 - expand_semantic_types_hierarchy={true|false} // default = false. true means to use the semantic types passed in the "semantic_types" parameter as well as all their immediate children. false means to use ONLY the semantic types passed in the "semantic_types" parameter.
 - expand_class_hierarchy={true|false} // default = false. used only in conjunction with "class_hierarchy_max_level" parameter; determines whether or not to include ancestors of the given class when performing an annotation.
 - class_hierarchy_max_level={0..N} // default = 0. the depth of the hierarchy to use when performing an annotation.
 - expand_mappings={true|false} // default = false. true means that the following manual mappings will be used in annotation: UMLS, REST, CUI, OBOXREF.
 - stop_words={word1,word2...,wordN} (case insensitive)
 - minimum_match_length={0..N}
 - exclude_numbers={true|false} // default = false
 - whole_word_only={true|false} // default = true
 - exclude_synonyms={true|false} // default = false
 - longest_only={true|false} // default = false. true means that only the longest match for a given phrase will be returned.
- Default stop words
 - The following stop words are used by default:
 - I, a, above, after, against, all, alone, always, am, amount, an, and, any, are, around, as, at, back, be, before, behind, below, between, bill, both, bottom, by, call, can, co, con, de, detail, do, done, down, due, during, each, eg, eight, eleven, empty, ever, every, few, fill, find, fire, first, five, for, former, four, from, front, full, further, get, give, go, had, has, hasnt, he, her, hers, him, his, i, ie, if, in, into, is, it, last, less, ltd, many, may, me, mill, mine, more, most, mostly, must, my, name, next, nine, no, none, nor, not, nothing, now, of, off, often, on, once, one, only, or, other, others, out, over, part, per, put, re, same, see, serious, several, she, show, side, since, six, so, some, sometimes, still, take, ten, the, then, third, this, thick, thin, three, through, to, together, top, toward, towards, twelve, two, un, under, until, up, upon, us, very, via, was, we, well, when, while, who, whole, will, with, within, without, you, yourself, yourselves

Recommender

The Recommender endpoint receives a text or a list of keywords and suggests appropriate ontologies for them.

- Recommend
 - GET POST /recommender?input={input text or keywords}
 - example: suggest ontologies for the text "Melanoma is a malignant tumor of melanocytes which are found predominantly in skin but also in the bowel and the eye": [/recommender?input=Melanoma%20is%20a%20malignant%20tumor%20of%20melanocytes%20which%20are%20found%20predominantly%20in%20skin%20but%20in%20the%20bowel%20and%20the%20eye](#)
- Parameters
 - input_type={1|2} // default = 1. 1 means that the input type is text. 2 means that the input type is a list of comma separated keywords.
 - output_type={1|2} // default = 1. 1 means that the output will be a ranked list of individual ontologies. 2 means that the output will be a ranked list of ontology sets.
 - max_elements_set={2|3|4} // default = 3. Maximum number of ontologies per set (only for output_type = 2).
 - wc={value in the range [0,1]} // default = 0.55. Weight assigned to the ontology coverage criterion.
 - wa={value in the range [0,1]} // default = 0.15. Weight assigned to the ontology acceptance criterion.
 - wd={value in the range [0,1]} // default = 0.15. Weight assigned to the ontology detail criterion.
 - ws={value in the range [0,1]} // default = 0.15. Weight assigned to the ontology specialization criterion.
 - ontologies={ontology_id1, ontology_id2, ..., ontology_idN} // default = (empty) (all BioPortal ontologies will be evaluated).
- Additional examples
 - Suggest ontologies for the **keywords** "leukocyte, phagocyte, motile cell, metabolising cell, dendritic cell, pipette": [/recommender?input=leukocyte,phagocyte,motile%20cell,metabolising%20cell,dendritic%20cell,%20pipette&input_type=2](#)
 - Suggest ontologies for the **keywords** "leukocyte, phagocyte, motile cell, metabolising cell, dendritic cell, pipette", **limited to the ontologies** UBERON, CL, NCIT and EFO: [/recommender?input=leukocyte,phagocyte,motile%20cell,metabolising%20cell,dendritic%20cell,%20pipette&input_type=2&ontologies=UBERON,CL,NCIT,EFO](#)
 - Suggest **ontology sets** for the **keywords** "leukocyte, phagocyte, motile cell, metabolising cell, dendritic cell, pipette": [/recommender?input=leukocyte,phagocyte,motile%20cell,metabolising%20cell,dendritic%20cell,%20pipette&input_type=2&output_type=2](#)
 - Suggest ontologies for the text "Melanoma is a malignant tumor of melanocytes which are found predominantly in skin but also in the bowel and the eye", **limiting the evaluation to ontology coverage** (the weights for the other criteria -acceptance, detail and specialization- will be set to 0): [/recommender?input=Melanoma%20is%20a%20malignant%20tumor%20of%20melanocytes%20which%20are%20found%20predominantly%20in%20skin%20but%20in%20the%20bowel%20and%20the%20eye&wc=0&wa=0&wd=0&ws=0](#)

Batch Endpoints

In order to reduce the number of HTTP requests required to obtain information about resources, we are exploring the use of batch endpoints that will use one request/response cycle for multiple resources. Currently, only classes are supported.

POST /batch

The batch service takes input in the BODY of a POST request. The input should be a JSON document with the following structure:

```
{
  "http://www.w3.org/2002/07/owl#Class": {
    "collection": [
      {
        "class": "http://example.org/ontology1/class1",
        "ontology": "http://example.org/ontology1"
      }
    ]
  }
}
```

```

    {
      "class": "http://example.org/ontology2/class1",
      "ontology": "http://example.org/ontology2"
    },
    {
      "class": "http://example.org/ontologyN/classN",
      "ontology": "http://example.org/ontologyN"
    }
  ],
  "display": "prefLabel,synonym,semanticTypes"
}
}

```

Example:

```

{
  "http://www.w3.org/2002/07/owl#Class": {
    "collection": [
      {
        "class": "http://bioontology.org/ontologies/BiomedicalResourceOntology.owl#Ontology_Development_and_Management",
        "ontology": "http://data.bioontology.org/ontologies/BR0"
      },
      {
        "class": "http://bioontology.org/ontologies/BiomedicalResourceOntology.owl#Modular_Component",
        "ontology": "http://data.bioontology.org/ontologies/BR0"
      },
      {
        "class": "http://bioontology.org/ontologies/BiomedicalResourceOntology.owl#Stimulator",
        "ontology": "http://data.bioontology.org/ontologies/BR0"
      }
    ],
    "display": "prefLabel,synonym,semanticTypes"
  }
}

```

Ontology Analytics

An endpoint for retrieving ontology visitor stats provided by Google Analytics.

Ontology Analytics by Month/Year

Allows to retrieve analytics for all ontologies for a specific month/year combination

- Get ALL analytics data
 - GET /analytics
 - example: [/analytics](#)
- Get analytics data for a given month/year
 - GET /analytics?month={month number}&year={year}
 - example: [/analytics?month=4&year=2014](#)
- Parameters
 - month={1-12}
 - year={2013-...}

Single Ontology Analytics

Allows to retrieve ALL analytics data for a single ontology

- Get analytics data for a given ontology
 - GET /ontologies/{acronym}/analytics
 - example: [/ontologies/NCIT/analytics](#)

Resource Endpoints

Endpoints for RESTful resources can be found below, specifically in the list of Media Types. Using the provided hypermedia links, you can navigate from resource to resource. Each Media Type has a corresponding collection URL (when available), a list of HTTP verbs that can be used to operate on the resources, a description of the resource (including available attributes), and a list of hypermedia links that can be found on each resource.

Media Types and Hypermedia Links

Documentation

The documentation below describes the media types that available in the API and the hypermedia links that connect them. Media types describe the types of resources available, including the HTTP verbs that may be used with them and the attributes that each resource contains.

Hypermedia Links

This programming interface comprises HTTP commands that return objects that themselves contain links to other locations in the form of a URL. In addition to the actual return values, the returned objects contain a set of links to related information. You can access these links using HTTP commands, typically GET. Thus, it is possible to traverse BioPortal's information for an ontology, including classes, notes, and reviews, by starting from the list of ontologies and traversing the available links - either programmatically or in a browser. The different resources are described below, including information about their available attributes, links, and the HTTP verbs that can be used to interact with them.

HTTP Verbs

The API uses different verbs to support processing of resources. This includes things like creating or deleting individual resources or something more specific searching or annotating. Typically, the verbs will be used in conjunction with the URL that represents the id for a given resource. Here is how we interpret the verbs:

- GET Used to retrieve a resource or collection of resources.
- POST Used to create a resource when the server determines the resource's id.
- PUT Used to create a resource when a client determines the resource's id.
- PATCH Used to modify an existing resource. The attributes in a PATCH request will replace existing attributes.
- DELETE Used to delete an existing resource.

Available Media Types

1. <http://data.bioontology.org/metadata/Category>
2. <http://www.w3.org/2002/07/owl#Class>
3. <http://data.bioontology.org/metadata/Group>
4. <http://www.w3.org/2002/07/owl#NamedIndividual>
5. <http://data.bioontology.org/metadata/Mapping>
6. <http://data.bioontology.org/metadata/Metrics>
7. <http://data.bioontology.org/metadata/Note>
8. <http://data.bioontology.org/metadata/Reply>
9. <http://www.w3.org/2002/07/owl#ObjectProperty>
10. <http://data.bioontology.org/metadata/Ontology>
11. <http://data.bioontology.org/metadata/OntologySubmission>
12. <http://data.bioontology.org/metadata/Project>
13. <http://data.bioontology.org/metadata/ProvisionalClass>
14. <http://data.bioontology.org/metadata/ProvisionalRelation>
15. <http://data.bioontology.org/metadata/Review>
16. <http://www.w3.org/2004/02/skos/core#Collection>
17. <http://www.w3.org/2004/02/skos/core#ConceptScheme>
18. <http://data.bioontology.org/metadata/Slice>
19. <http://data.bioontology.org/metadata/User>

http://data.bioontology.org/metadata/Category

Resource collection: </categories>

HTTP Methods for Resource

HTTP Verb Path

GET	/ontologies/:acronym/categories
GET	/categories
GET	/categories/{acronym}
PUT	/categories/{acronym}
POST	/categories
PATCH	/categories/{acronym}
DELETE	/categories/{acronym}

Resource Description

Attribute	Default	Unique	Required	List	Type
acronym	true	true	true		
name	true		true		
description	true				
created	true				
parentCategory	true				http://data.bioontology.org/metadata/Category
ontologies	true				http://data.bioontology.org/metadata/Ontology

http://www.w3.org/2002/07/owl#Class

Example:

[/ontologies/SNOMEDCT/classes/http%3A%2F%2Fpurl.bioontology.org%2Fontology%2FSNOMEDCT%2F410607006](http://ontologies/SNOMEDCT/classes/http%3A%2F%2Fpurl.bioontology.org%2Fontology%2FSNOMEDCT%2F410607006)

HTTP Methods for Resource

HTTP Verb Path

GET	/ontologies/:ontology/classes
GET	/ontologies/{ontology}/classes/roots_paged
GET	/ontologies/{ontology}/classes/roots
GET	/ontologies/{ontology}/classes/{cls}
GET	/ontologies/{ontology}/classes/{cls}/paths_to_root
GET	/ontologies/{ontology}/classes/{cls}/tree
GET	/ontologies/{ontology}/classes/{cls}/ancestors
GET	/ontologies/{ontology}/classes/{cls}/descendants
GET	/ontologies/{ontology}/classes/{cls}/children
GET	/ontologies/{ontology}/classes/{cls}/parents

Resource Description

Attribute	Default	Unique	Required	List	Type
prefLabel	true		true		
synonym	true			true	
definition	true			true	
obsolete	true				
semanticType	true			true	
cui	true			true	
inScheme	true			true	
memberOf	true				http://www.w3.org/2004/02/skos/core#Collection
created	true				
modified	true				
label				true	
prefLabelXI				true	http://www.w3.org/2008/05/skos-xl#Label
altLabelXI				true	http://www.w3.org/2008/05/skos-xl#Label
hiddenLabelXI				true	http://www.w3.org/2008/05/skos-xl#Label
notation					
prefixIRI					
parents				true	http://www.w3.org/2002/07/owl#Class
ancestors					http://www.w3.org/2002/07/owl#Class
children					http://www.w3.org/2002/07/owl#Class
subClassOf				true	
xref					
notes					http://data.bioontology.org/metadata/Note
properties					
childrenCount					
hasChildren					

Related Hypermedia Links

Type	URL
self	https://data.bioontology.org/ontologies/BRO/classes/http%3A%2F%2Fbioontology.org%2Fontologies%2FBiomedicalResourceOntology.owl%3Fontology%3DBRO
ontology	https://data.bioontology.org/ontologies/BRO
children	https://data.bioontology.org/ontologies/BRO/classes/http%3A%2F%2Fbioontology.org%2Fontologies%2FBiomedicalResourceOntology.owl%3Fchildren
parents	https://data.bioontology.org/ontologies/BRO/classes/http%3A%2F%2Fbioontology.org%2Fontologies%2FBiomedicalResourceOntology.owl%3Fparents
descendants	https://data.bioontology.org/ontologies/BRO/classes/http%3A%2F%2Fbioontology.org%2Fontologies%2FBiomedicalResourceOntology.owl%3Fdescendants
ancestors	https://data.bioontology.org/ontologies/BRO/classes/http%3A%2F%2Fbioontology.org%2Fontologies%2FBiomedicalResourceOntology.owl%3Fancestors

<http://data.bioontology.org/metadata/Group>

HTTP Methods for Resource

Attribute	Default	Unique	Required	List	Type
acronym	true	true	true		
name	true		true		
description	true				
created	true				
ontologies	true				http://www.semanticweb.org/ontologies/2008/05/ontology-11.owl

<http://www.w3.org/2002/07/owl#NamedIndividual>

HTTP Methods for Resource

Resource Description

<http://data.bioontology.org/metadata/Mapping>

Example:

[/ontologies/SNOMEDCT/classes/http%3A%2F%2Fpurl.bioontology.org%2Fontology%2FSNOMEDCT%2F410607006/mappings](http://ontologies/SNOMEDCT/classes/http%3A%2F%2Fpurl.bioontology.org%2Fontology%2FSNOMEDCT%2F410607006/mappings)

HTTP Methods for Resource

HTTP Verb Path

GET	/ontologies/:ontology/classes/:cls/mappings
GET	/ontologies/:ontology/mappings
GET	/mappings
GET	/mappings/recent
GET	/mappings/{mapping}
GET	/mappings/statistics/ontologies
GET	/mappings/statistics/ontologies/{ontology}
POST	/mappings
DELETE	/mappings/{mapping}

Resource Description

Attribute Default Unique Required List Type

process
classes
id
source

<http://data.bioontology.org/metadata/Metrics>

Resource collection: [/metrics](#)

HTTP Methods for Resource

HTTP Verb Path

GET	/metrics
GET	/metrics/missing
GET	/ontologies/:ontology/metrics
GET	/ontologies/:ontology/submissions/:ontology_submission_id/metrics

Resource Description

Attribute	Default	Unique	Required	List	Type
submission	true				http://data.bioontology.org/metadata/OntologySubmission
created	true				
classes	true		true		
individuals	true		true		
properties	true		true		
maxDepth	true		true		
maxChildCount	true		true		
averageChildCount	true		true		
classesWithOneChild	true		true		
classesWithMoreThan25Children	true		true		
classesWithNoDefinition	true		true		

<http://data.bioontology.org/metadata/Note>

Example: </ontologies/NCIT/notes>

HTTP Methods for Resource

HTTP Verb Path

GET	/ontologies/:ontology/notes
GET	/ontologies/:ontology/classes/:cls/notes
GET	/notes
GET	/notes/{noteid}
POST	/notes
PATCH	/notes/{noteid}
DELETE	/notes/{noteid}

Resource Description

Attribute	Default	Unique	Required	List	Type
subject	true				
body	true				
creator	true		true		http://data.bioontology.org/metadata/User
created	true				
archived	true				
createdInSubmission	true				http://data.bioontology.org/metadata/OntologySubmission
reply	true			true	http://data.bioontology.org/metadata/Reply
relatedOntology	true		true	true	http://data.bioontology.org/metadata/Ontology
relatedClass	true			true	http://www.w3.org/2002/07/owl#Class
proposal	true				http://data.bioontology.org/metadata/Base

<http://data.bioontology.org/metadata/Reply>

Resource collection: </replies>

HTTP Methods for Resource

HTTP Verb Path

GET	/notes/:noteid/replies
GET	/replies
GET	/replies/{replyid}
POST	/replies
PATCH	/replies/{replyid}
DELETE	/replies/{replyid}

Resource Description

Attribute	Default	Unique	Required	List	Type
body	true	true			
creator	true		true		http://data.bioontology.org/metadata/User
created	true				
children	true				http://data.bioontology.org/metadata/Reply
parent					http://data.bioontology.org/metadata/Reply

<http://www.w3.org/2002/07/owl#ObjectProperty>

Example: coming soon

HTTP Methods for Resource**HTTP Verb Path**

GET	/ontologies/ontology/properties
GET	/ontologies/{ontology}/properties/roots
GET	/ontologies/{ontology}/properties/{property}
GET	/ontologies/{ontology}/properties/{property}/label
GET	/ontologies/{ontology}/properties/{property}/tree
GET	/ontologies/{ontology}/properties/{property}/ancestors
GET	/ontologies/{ontology}/properties/{property}/descendants
GET	/ontologies/{ontology}/properties/{property}/parents
GET	/ontologies/{ontology}/properties/{property}/children

Resource Description

Attribute	Default	Unique	Required	List	Type
label	true			true	
definition	true			true	
parents	true			true	http://www.w3.org/2002/07/owl#ObjectProperty
children	true				http://www.w3.org/2002/07/owl#ObjectProperty
domain	true				
range	true				
submission					
ancestors					
descendants					
properties					

<http://data.bioontology.org/metadata/Ontology>Resource collection: [/ontologies](#)**HTTP Methods for Resource****HTTP Verb Path**

GET	/ontologies
GET	/ontologies/{acronym}
GET	/ontologies/{acronym}/latest_submission
GET	/ontologies/{acronym}/download
GET	/ontologies/{acronym}/admin/log
GET	/ontologies_full
PUT	/ontologies/{acronym}
POST	/ontologies/{acronym}/pull
POST	/ontologies
PATCH	/ontologies/{acronym}
DELETE	/ontologies/{acronym}

Resource Description

Attribute	Default	Unique	Required	List	Type
acronym	true	true	true		
name	true	true	true		
administeredBy	true		true	true	http://data.bioontology.org/metadata/User

flat	true	
summaryOnly	true	
ontologyType	true	http://data.bioontology.org/metadata/OntologyType
submissions		http://data.bioontology.org/metadata/OntologySubmission
projects		http://data.bioontology.org/metadata/Project
notes		http://data.bioontology.org/metadata/Note
reviews		http://data.bioontology.org/metadata/Review
provisionalClasses		http://data.bioontology.org/metadata/ProvisionalClass
subscriptions		http://data.bioontology.org/metadata/Subscription
group	true	http://data.bioontology.org/metadata/Group
viewingRestriction		
doNotUpdate		
hasDomain	true	http://data.bioontology.org/metadata/Category
acl	true	http://data.bioontology.org/metadata/User
viewOf		http://data.bioontology.org/metadata/Ontology
views		http://data.bioontology.org/metadata/Ontology

Related Hypermedia Links

Type	URL
submissions	https://data.bioontology.org/ontologies/BRO/submissions
properties	https://data.bioontology.org/ontologies/BRO/properties
classes	https://data.bioontology.org/ontologies/BRO/classes
single_class	https://data.bioontology.org/ontologies/BRO/classes/{class_id}
roots	https://data.bioontology.org/ontologies/BRO/classes/roots
schemes	https://data.bioontology.org/ontologies/BRO/schemes
collections	https://data.bioontology.org/ontologies/BRO/collections
xl_labels	https://data.bioontology.org/ontologies/BRO/skos_xl_labels
instances	https://data.bioontology.org/ontologies/BRO/instances
metrics	https://data.bioontology.org/ontologies/BRO/metrics
reviews	https://data.bioontology.org/ontologies/BRO/reviews
notes	https://data.bioontology.org/ontologies/BRO/notes
groups	https://data.bioontology.org/ontologies/BRO/groups
categories	https://data.bioontology.org/ontologies/BRO/categories
latest_submission	https://data.bioontology.org/ontologies/BRO/latest_submission
projects	https://data.bioontology.org/ontologies/BRO/projects
download	https://data.bioontology.org/ontologies/BRO/download
views	https://data.bioontology.org/ontologies/BRO/views
analytics	https://data.bioontology.org/ontologies/BRO/analytics
ui	http://biportal.bioontology.org/ontologies/BRO

<http://data.bioontology.org/metadata/OntologySubmission>

Example: [/ontologies/NCIT/submissions?display=submissionId,version](https://data.bioontology.org/ontologies/NCIT/submissions?display=submissionId,version)

HTTP Methods for Resource

HTTP Verb	Path
GET	/submissions
GET	/ontologies/{acronym}/submissions
GET	/ontologies/{acronym}/submissions/{ontology_submission_id}

GET	/ontologies/{acronym}/submissions/bulk_delete/{process_id}
GET	/ontologies/{acronym}/submissions/{ontology_submission_id}/download
GET	/ontologies/{acronym}/submissions/{ontology_submission_id}/download_diff
POST	/submissions
POST	/ontologies/:acronym/submissions
PATCH	/ontologies/{acronym}/submissions/{ontology_submission_id}
DELETE	/ontologies/{acronym}/submissions/{ontology_submission_id}
DELETE	/ontologies/:acronym/submissions

Resource Description

Attribute	Default	Unique	Required	List	Type
submissionId	true		true		
version	true				
status	true				
hasOntologyLanguage	true		true		http://data.bioontology.org/metadata/OntologyFormat
description	true				
homepage	true				
documentation	true				
publication	true			true	
released	true		true		
creationDate	true				
contact	true		true	true	http://data.bioontology.org/metadata/Contact
ontology	true		true		http://data.bioontology.org/metadata/Ontology
prefLabelProperty					
definitionProperty					
synonymProperty					
authorProperty					
classType					
hierarchyProperty					
obsoleteProperty					
obsoleteParent					
createdProperty					
modifiedProperty					
versionIRI					
deprecated					
hasFormalityLevel					
hasOntologySyntax					
naturalLanguage				true	
isOfType					
identifier				true	
uri					
notes				true	
keywords				true	
hiddenLabel				true	
alternative				true	
abstract					
hasLicense					
useGuidelines					
morePermissions					

valid	
curatedOn	true
modificationDate	
audience	
repository	
bugDatabase	
mailingList	
toDoList	true
award	true
knownUsage	true
designedForOntologyTask	true
hasDomain	true
coverage	
example	true
conformsToKnowledgeRepresentationParadigm	
usedOntologyEngineeringMethodology	
usedOntologyEngineeringTool	true
accrualMethod	true
accrualPeriodicity	
accrualPolicy	
competencyQuestion	true
wasGeneratedBy	true
wasInvalidatedBy	true
pullLocation	
isFormatOf	
hasFormat	true
dataDump	
csvDump	
uriLookupEndpoint	
openSearchDescription	
source	true
endpoint	true
includedInDataCatalog	true
hasPriorVersion	
hasPart	true
ontologyRelatedTo	true
similarTo	true
comesFromTheSameDomain	true
isAlignedTo	true
isBackwardCompatibleWith	true
isIncompatibleWith	true
hasDisparateModelling	true
hasDisjunctionsWith	true
generalizes	true
explanationEvolution	true
useImports	true
usedBy	true
workTranslation	true
translationOfWork	true

uriRegexPattern	
preferredNamespaceUri	
preferredNamespacePrefix	
exampleIdentifier	
keyClasses	true
metadataVoc	true
uploadFilePath	
diffFilePath	
masterFileName	
associatedMedia	true
depiction	true
logo	
metrics	http://data.bioontology.org/metadata/Metrics
submissionStatus	true http://data.bioontology.org/metadata/SubmissionStatus
missingImports	true

http://data.bioontology.org/metadata/Project

Resource collection: [/projects](#)

HTTP Methods for Resource

HTTP Verb	Path
GET	/ontologies/:acronym/projects
GET	/projects
GET	/projects/{acronym}
PUT	/projects/{acronym}
POST	/projects
PATCH	/projects/{acronym}
DELETE	/projects/{acronym}

Resource Description

Attribute	Default	Unique	Required	List	Type
acronym	true	true	true		
creator	true		true	true	http://data.bioontology.org/metadata/User
created	true				
updated	true				
name	true		true		
homePage	true		true		
description	true		true		
contacts	true				
institution	true				
ontologyUsed	true			true	http://data.bioontology.org/metadata/Ontology

http://data.bioontology.org/metadata/ProvisionalClass

Resource collection: [/provisional_classes](#)

HTTP Methods for Resource

HTTP Verb	Path
-----------	------

GET	/ontologies/:ontology/provisional_classes
GET	/users/:user/provisional_classes
GET	/provisional_classes
GET	/provisional_classes/{provisional_class_id}
POST	/provisional_classes
PATCH	/provisional_classes/{provisional_class_id}
DELETE	/provisional_classes/{provisional_class_id}

Resource Description

Attribute	Default	Unique	Required	List	Type
label	true		true		
synonym	true			true	
definition	true			true	
subclassOf	true				
creator	true		true		http://data.bioontology.org/metadata/User
created	true				
permanentId	true				
notId	true				
ontology	true				http://data.bioontology.org/metadata/Ontology
relations	true				http://data.bioontology.org/metadata/ProvisionalRelation

http://data.bioontology.org/metadata/ProvisionalRelation

Resource collection: /provisional_relations

HTTP Methods for Resource**HTTP Verb Path**

GET	/provisional_relations
GET	/provisional_relations/{provisional_relation_id}
POST	/provisional_relations
DELETE	/provisional_relations/{provisional_relation_id}

Resource Description

Attribute	Default	Unique	Required	List	Type
source	true		true		http://data.bioontology.org/metadata/ProvisionalClass
relationType	true		true		
targetClassId	true		true		
targetClassOntology	true		true		http://data.bioontology.org/metadata/Ontology
creator	true		true		http://data.bioontology.org/metadata/User
created	true				

http://data.bioontology.org/metadata/Review

Resource collection: </reviews>

HTTP Methods for Resource**HTTP Verb Path**

GET	/ontologies/:acronym/reviews
GET	/reviews

GET	/reviews/{review_id}
POST	/reviews
PATCH	/reviews/{review_id}
DELETE	/reviews/{review_id}

Resource Description

Attribute	Default	Unique	Required	List	Type
creator	true		true		http://data.bioontology.org/metadata/User
created	true				
updated	true				
body	true		true		
ontologyReviewed	true		true		http://data.bioontology.org/metadata/Ontology
usabilityRating	true				
coverageRating	true				
qualityRating	true				
formalityRating	true				
correctnessRating	true				
documentationRating	true				

<http://www.w3.org/2004/02/skos/core#Collection>

Example: coming soon

HTTP Methods for Resource**HTTP Verb Path**

GET	/ontologies/:ontology/collections
GET	/ontologies/{ontology}/collections/{collection}
GET	/ontologies/{ontology}/collections/{collection}/members

Resource Description

Attribute	Default	Unique	Required	List	Type
prefLabel	true		true		
memberCount	true				
properties					

<http://www.w3.org/2004/02/skos/core#ConceptScheme>

Example: coming soon

HTTP Methods for Resource**HTTP Verb Path**

GET	/ontologies/:ontology/schemes
GET	/ontologies/{ontology}/schemes/{scheme}

Resource Description

Attribute	Default	Unique	Required	List	Type
prefLabel	true		true		
properties	true				

http://data.bioontology.org/metadata/SliceResource collection: </slices>**HTTP Methods for Resource****HTTP Verb Path**

GET	/slices
GET	/slices/{slice_id}
GET	/slices/synchronize_groups
POST	/slices
PATCH	/slices/{slice}
DELETE	/slices/{slice}

Resource Description

Attribute	Default	Unique	Required	List	Type
acronym	true	true	true		
name	true		true		
description	true				
created	true				
ontologies	true		true	true	http://data.bioontology.org/metadata/Ontology

http://data.bioontology.org/metadata/UserResource collection: </users>**HTTP Methods for Resource****HTTP Verb Path**

GET	/users
GET	/users/{username}
GET	/users/{username}/ontologies
PUT	/users/{username}
POST	/users/authenticate
POST	/users/create_reset_password_token
POST	/users/reset_password
POST	/users
PATCH	/users/{username}
DELETE	/users/{username}

Resource Description

Attribute	Default	Unique	Required	List	Type
username	true	true	true		
email	true		true		
role	true			true	http://data.bioontology.org/metadata/Role
apikey	true	true			
firstName					
lastName					
subscribed					
githubId		true			
orcidId		true			

created	
lastLoginAt	
subscription	true http://data.bioontology.org/metadata/Subscription
customOntology	true http://data.bioontology.org/metadata/Ontology
provisionalClasses	http://data.bioontology.org/metadata/ProvisionalClass
createdOntologies	true

Content Types

The API returns JSON as the default content type. This can be overridden by using the `format` query string parameter with the value `json`, `jsonp`, or `xml`. The API also respects `Accept` header entries (EX: `application/json`, `application/xml`), with precedence given to the `format` parameter.

JSON

The default content type is JSON, specifically a variant called [JSON-LD](#), or JSON Linked Data. You can treat this variant like normal JSON. All JSON parsers will be able to parse the output normally. The benefit of JSON-LD is that it enables hypermedia links, and you will find these links exposed as URLs in attributes labeled `@id`, which correspond to the id of the parent resource, or in an array called `links`, which contains a hash of link types with corresponding URLs.

Here is a sample output of the JSON response format:

```
{
  "administeredBy": [
    "http://data.bioontology.org/user/nevada"
  ],
  "acronym": "ABA-API-TST",
  "name": "ABA Adult Mouse Brain",
  "@id": "http://data.bioontology.org/ontology/ABA-API-TST",
  "@type": "http://data.bioontology.org/metadata/Ontology",
  "links": {
    "metrics": "http://data.bioontology.org/ontologies/ABA-API-TST/metrics",
    "submissions": "http://data.bioontology.org/ontologies/ABA-API-TST/submissions",
    "classes": "http://data.bioontology.org/ontologies/ABA-API-TST/classes",
    "roots": "http://data.bioontology.org/ontologies/ABA-API-TST/classes/roots",
    "reviews": "http://data.bioontology.org/ontologies/ABA-API-TST/reviews"
  },
  "@context": {
    "@vocab": "http://data.bioontology.org/metadata/",
    "acronym": "http://omv.ontoware.org/2005/05/ontology#acronym",
    "name": "http://omv.ontoware.org/2005/05/ontology#name",
    "administeredBy": {
      "@id": "http://data.bioontology.org/metadata/User",
      "@type": "@id"
    }
  }
}
```

- Line 7 shows the id for the resource. Doing an HTTP GET on the id will retrieve the resource.
- Line 8 shows the media type (see below).
- Line 9 starts the links hash.
- Line 16 is the resource's context, which can be used to determine the type for lists of ids. For example, line 2 lists the ids for users who administer the ontology, which can be determined by looking for the `administeredBy` attribute in the `@context` hash.
- If you are interested in the predicate URI values used in the resource, these can be determined by looking up the attribute in the `@context` hash or by appending the value of `@vocab` (line 17) to an attribute name in cases where the attribute isn't listed specifically in the `@context`.

XML

XML is also available as an alternative content type.

Here is sample output for the XML format:

```
<ontology>
  <administeredByCollection>
    <administeredBy>http://data.bioontology.org/user/nevada</administeredBy>
  </administeredByCollection>
  <acronym>ABA-API-TST</acronym>
  <name>ABA Adult Mouse Brain</name>
  <id>http://data.bioontology.org/ontology/ABA-API-TST</id>
  <links>
    <self href="http://data.bioontology.org/ontology/ABA-API-TST" rel="http://data.bioontology.org/metadata/Ontology"/>
    <metrics href="/ontologies/ABA-API-TST/metrics"/>
    <submissions href="/ontologies/ABA-API-TST/submissions" rel="http://data.bioontology.org/metadata/OntologySubmission"/>
    <classes href="/ontologies/ABA-API-TST/classes" rel="http://www.w3.org/2002/07/owl#Class"/>
    <roots href="/ontologies/ABA-API-TST/classes/roots" rel="http://www.w3.org/2002/07/owl#Class"/>
    <reviews href="/ontologies/ABA-API-TST/reviews" rel="http://data.bioontology.org/metadata/Review"/>
  </links>
</ontology>
```

- Line 8 starts the links section
- Lines 9-14 list links by type. The `href` attribute contains the link location and the `rel` attribute defines the type of resource that will be found at that location.
- Elements outside of the links can also contain `href` and `rel` attributes (coming soon...)